

Characterizing Consensus Protocols in Heterogeneous Clusters

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Abstract

This paper reports a systematic evaluation of distributed cache hierarchies. The measurements reported here were collected over twelve independent trials under controlled conditions. Residual error grows approximately linearly with the number of coupled parameters. Calibration drift over the trial period remained below the threshold set out in the protocol.

1. Introduction

Sensitivity to the smoothing window proved modest across the range we examined. Taken together, these findings argue for treating the two regimes separately in future analyses. The measurements reported here were collected over twelve independent trials under controlled conditions. Neither of the alternative formulations improved on the simple additive model in our setting. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. The reduced model retains most of the explanatory power at a fraction of the computational cost. Residual error grows approximately linearly with the number of coupled parameters.

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The reduced model retains most of the explanatory power at a fraction of the computational cost. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity.

2. Related Work

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3. Materials and Methods

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4. Experimental Setup

We release the processing scripts and intermediate artefacts to support independent verification. The reduced model retains most of the explanatory power at a fraction of the computational cost. The measurements reported here were collected over twelve independent trials under controlled conditions. Neither of the alternative formulations improved on the simple additive model in our setting. Residual error grows approximately linearly with the number of coupled parameters. Sensitivity to the smoothing window proved modest across the range we examined.

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5. Results

Neither of the alternative formulations improved on the simple additive model in our setting. Replicate agreement was high, suggesting the protocol is robust to small handling differences. The measurements reported here were collected over twelve independent trials under controlled conditions. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. The reduced model retains most of the explanatory power at a fraction of the computational cost. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. We release the processing scripts and intermediate artefacts to support independent verification. Sensitivity to the smoothing window proved modest across the range we examined.

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6. Discussion

A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Neither of the alternative formulations improved on the simple additive model in our setting. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Longer observation windows narrow the confidence interval but do not shift the central estimate. Taken together, these findings argue for treating the two regimes separately in future analyses. Sensitivity to the smoothing window proved modest across the range we examined.

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7. Limitations

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8. Conclusion

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